DELHI PUBLIC SCHOOL JANKIPURAM

SESSION 2024 - 2025 : UNIT TEST 1 MATHEMATICS

Max. Marks: 20 Time: 45 min

Date: 22/5/24 Name: Mi Crocus Class IX Sec D

General Instructions:

- **All questions are compulsory.**
- **The question paper consists of 11 questions divided into 4 sections,**
- **Section A contains 7 questions of 1 mark each. Section B contains 2 questions of 2 marks each. Section C contains 1 question of 5 marks and Section D contains 1 case study question of 4 marks. Internal choice has been provided in Section B and C for one question.**
- **This question paper consists of 2 pages.**

SECTION A (1 mark)

1) Which one of the following is a polynomial?

a) $\frac{x^2}{3} - \frac{2}{x^2}$ b) $x^3 + \frac{4x^{3/2}}{\sqrt{x}}$

c) $\sqrt{3y} + 5$ d) $\frac{x^2-1}{x^2+1}$

2) If we add or subtract two irrational numbers, we get _____ number.
(a) natural (b) rational
(c) irrational (d) rational or irrational

3) A polynomial of degree 5 in x has at most:
(a) 5 terms (b) 10 terms (c) 6 terms (d) 4 terms

4) By simplifying $(7 + \sqrt{5})(7 - \sqrt{5})$, we get 44
(a) 24 (b) 2 (c) 12 (d) 74

5) The degree of the polynomial $\sqrt{2} - y - \sqrt{3}\, y^2 + y^7$
(a) 7 (b) 2 (c) 1 (d) 9

6) **Assertion:** $\sqrt{5}$ is an irrational number.
Reason: A number is called irrational, if it cannot be written in the form p/q, where p and q are integers and $q \neq 0$
a.) Both Assertion and Reason are correct and Reason is the correct explanation for Assertion
b.) Both Assertion and Reason are correct and Reason is not the correct explanation for Assertion.
c.) Assertion is true but the reason is false.
d.) Both assertion and reason are false.

7) **Assertion:** degree of non-zero constant polynomial is zero
Reason: polynomial having two terms are called binomial.
a.) Both Assertion and Reason are correct and Reason is the correct explanation for Assertion
b.) Both Assertion and Reason are correct and Reason is not the correct explanation for Assertion.
c.) Assertion is true but the reason is false.
d.) Both assertion and reason are false.

SECTION B (2 marks)

8) If $p(x) = x^2 - 2\sqrt{2}x + 1$, then find the value of $p(\sqrt{2})$

9) Express $23.\overline{43}$ in p/q form, where p and q are integers and $q \neq 0$.

Or

Insert four rational numbers between $\frac{1}{3}$ and $\frac{2}{5}$.

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SECTION C (5 marks)

10) Represent $\sqrt{5}$ on number line by choosing proper scale.

Or

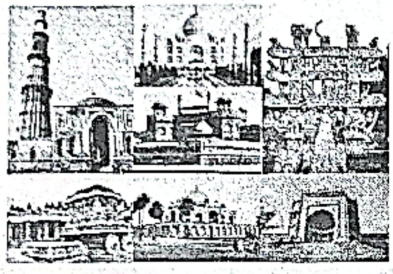
Find a and b if:

$$\frac{3-\sqrt{6}}{3+2\sqrt{6}} = a\sqrt{6} - b$$

SECTION D (4 marks)

CASE STUDY - I:

11) A Boarding School of Lucknow decided different types of tours for the students to educate them. So, in class 9th, $\frac{1}{12}th$ times the square of the total number of students planned to visit historical monuments. $\frac{7}{12}th$ Times the member of students planned to visit old age homes while 15 students decided to teach poor children.







i. Using above information, express the total number of students as a polynomial p(x) in term of x. (1)

a) $\frac{1}{12}x^2 + \frac{7}{12}x + 15$

c) $\frac{1}{4}x^2 + \frac{7}{4}x + 10$

b) $\frac{7}{12}x^2 + \frac{1}{12}x + 15$

d) $\frac{1}{12}x^2 + \frac{7}{12}x + 15 + x$

ii. Write the coefficient of x^0 in polynomial p(x). (1)

a) $\frac{7}{4}$ b)15 c) $\frac{1}{12}$ d)10

iii. Write the degree polynomial p(x) 2 (1)

iv. The value of $p(x)$ at $x = 2$ is $\frac{33}{2}$ (1)

$\frac{33}{2}$

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